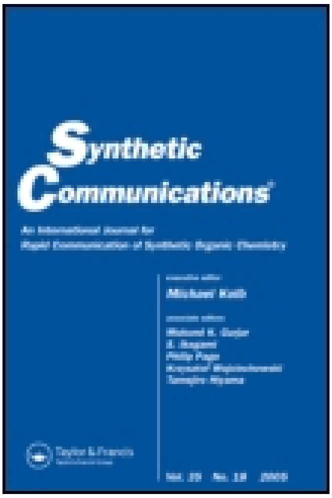

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New C-Glycosylated α-Amino Acid Synthesis by Addition Reaction of an Amino Acid Organozinc Reagent on Carbohydrate-Derived Aldehydes

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NEW *C*-GLYCOSYLATED α-AMINO ACID SYNTHESIS BY ADDITION REACTION OF AN AMINO ACID ORGANOZINC REAGENT ON CARBOHYDRATE-DERIVED ALDEHYDES

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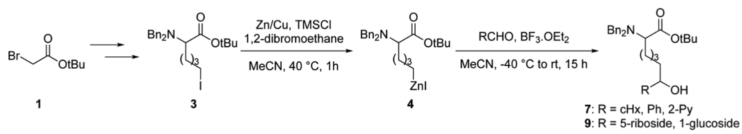
GRAPHICAL ABSTRACT



Abstract *Two unreported C-glycosylated amino acid structures could be prepared by addition of an amino acid organozinc on carbohydrates bearing an aldehyde function. For this purpose, a new iodinated amino acid bearing a four-carbon side chain was conveniently prepared in two steps. Conversion into the corresponding organozinc reagent was optimal in acetonitrile using zinc/copper activated with 1,2-dibromoethane and trimethylsilyl chloride. A protocol was developed for the addition of this organozinc compound to simple aldehydes in acetonitrile using $BF_3 \cdot OEt_2$ as a Lewis acid. When applied to riboside and glucoside aldehyde derivatives, this protocol led to new C-glycosylated amino acids in acceptable yields.*

Keywords Amino acid; *C*-glycosylated amino acid; organozinc

INTRODUCTION

Glycoproteins and glycopeptides are molecules bearing a carbohydrate and a peptidic domain linked by glycosidic bonds. In most cases, those bonds are *N*-glycosidic (through an asparagine residue) or *O*-glycosidic (through serine, threonine, or tyrosine residues).[1] Glycoproteins are present in every living organisms and play important roles in biological events such as intercellular communication, molecular recognition, microbial virulence,[2] inflammation processes,[3] and immune responses.[4] However, *N*- and *O*-glycosidic linkages are sensitive to enzymatic

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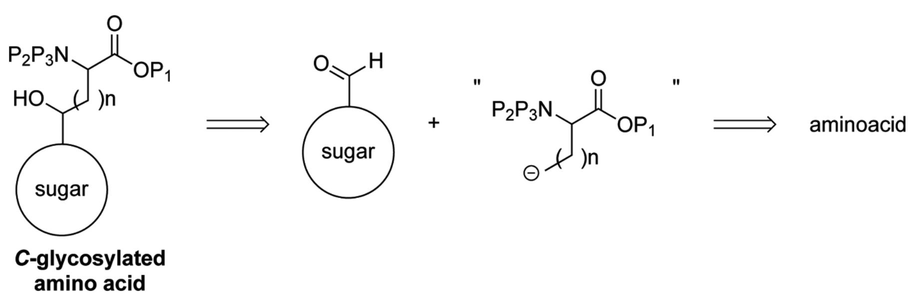
cleavages by glycosidases, or to acidic and basic media, complicating the synthesis and biological evaluation of glycopeptidic drugs.

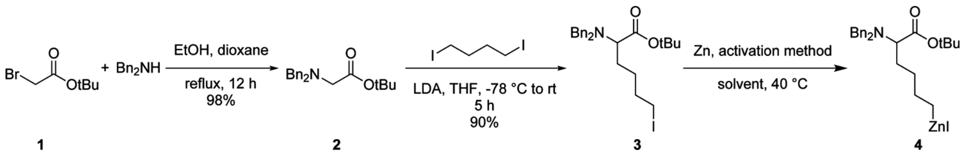
Over the past 20 years, increasing efforts have been involved in the synthesis of amino acids linked by their α-carbon to carbohydrate units by carbon–carbon bonds, either directly or through a carbon tether. Synthetic approaches used for the synthesis of these compounds cover a wide range of carbon–carbon bond-forming methodologies (phosphorus and sulfur ylide chemistry, use of metallated nucleophiles, radical chemistry, Strecker reaction, Claisen rearrangement, Ramberg–Bäcklund reaction, or palladium cross coupling) and have been extensively reviewed.[5] Those more stable *C*-glycosylated amino acids could lead to drugs with improved pharmacokinetic properties.[6]

As part of our ongoing program on new *C*-glycosylated amino acid syntheses, our group already showed that this type of compounds could be obtained by addition of glycine enolates on sugars bearing a carbonyl group.[7] Our research efforts led us to consider the possibility of installing a carbon tether between the amino acid and the carbohydrate moieties. This could be achieved by addition of a suitable carbanionic nucleophile equivalent, located at the end of the side chain of an amino acid, on an electrophilic function, such as an aldehyde, localized on the carbohydrate moiety (Scheme 1).

Protected lithiated alaninol prepared from serine has been shown to react on electrophiles such as aldehydes and ketones to give addition products in good yields.[8] However, this approach required subsequent deprotection and reoxidation of the alcohol function to recover the amino acid functionality of the target compounds. In spite of their low reactivity, organozinc reagents display a high selectivity, especially in the presence of highly functionalized substrates.[9] Amino acid organozinc compounds, prepared by zinc insertion reaction on amino esters iodinated on their side chain, have been extensively studied by Jackson et al.[10] These reagents could be synthesized and used with conservation of the carboxyl functionality. This group reported a synthesis of *C*-glycosylated amino acids by conjugate S_N2' reaction of various zinc/copper reagents derived from amino acids on glucals bearing a leaving group on position 3.[11] Stimulated by those results, we decided to explore the unprecedented addition of amino acid organozinc reagents on aldehydes for its application to *C*-glycosylated amino acid synthesis.



Scheme 1. Synthetic strategy.



Scheme 2. Synthesis of amino acid organozinc **4**.

We report herein a proof of concept study concerning the preparation of amino acid organozinc reagent **4** (Scheme 2) and its subsequent "one-pot" addition on aldehydes, more particularly on riboside and glucoside derivatives bearing an aldehyde function, leading to new *C*-glycosyl amino acid structures.

RESULTS AND DISCUSSION

To explore amino acid organozinc addition on aldehydes, we decided to use *tert*-butyl 2-(dibenzylamino)-6-iodohexanoate **3** (Scheme 2) as a readily available iodinated precursor. This previously unreported iodinated derivative was prepared from *tert*-butyl bromoacetate **1** in a two step sequence consisting in reaction with dibenzylamine[12] and alkylation of the resulting glycine derivative **2** with 1,4-diiodobutane after conversion into its lithium enolate with lithium diisopropylamide (LDA, Scheme 2).[13] LDA was prepared *in situ* by reaction of diisopropylamine and butyllithium. Better alkylation yields were obtained using butyllithium prepared in diethyl ether instead of commercially available hexane solutions. The use of an excess of 1,4-diiodobutane (4 equiv.) reduced the formation of the double substitution by-product. Iodinated precursor **3** was obtained in good overall yield (88%) and could be prepared on a large scale (up to 20 mmol). Attempts to use 1,3-diiodopropane or 1,2-diiodoethane to access shorter iodinated amino acid analogs were unsuccessful. In the case of 1,3-diiodopropane, amine quaternarization occurred instantaneously after alkylation, leading exclusively to the favored five-member cyclic salt. Interestingly, such reaction was not observed on compound **3** even after prolonged storage periods. The use of 1,2-diiodoethane invariably yielded mixtures of unchanged starting material and vinylglycine arising from LDA-mediated elimination after alkylation. The use of longer linear diiodoalkanes (>C_4) was not considered because it would lead to compounds losing bio-isosterism with natural *N*- and *O*-glycosylated amino acids.

Formation of the organozinc reagent **4** depended on the zinc activation method[14] and the solvent employed. Different procedures for the activation of zinc powder were first examined in dry tetrahydrofuran (THF) (Table 1) before the zincation reaction of compound **3** was performed at 40 °C. The reaction was monitored by liquid chromatography–mass spectroscopy (LC-MS) analysis of an aliquot hydrolyzed with D_2O. Conversion was calculated as the ratio between starting iodinated material and hydrolyzed product. The ratio of protonated versus deuterated compound shown by MS analysis allowed us to evaluate the effective quantity of organozinc compound in the reaction mixture.

The use of catalytic quantities of iodine crystals or 1,2-dibromoethane (10 mol%) as activating agents failed to give the expected organozinc compound.

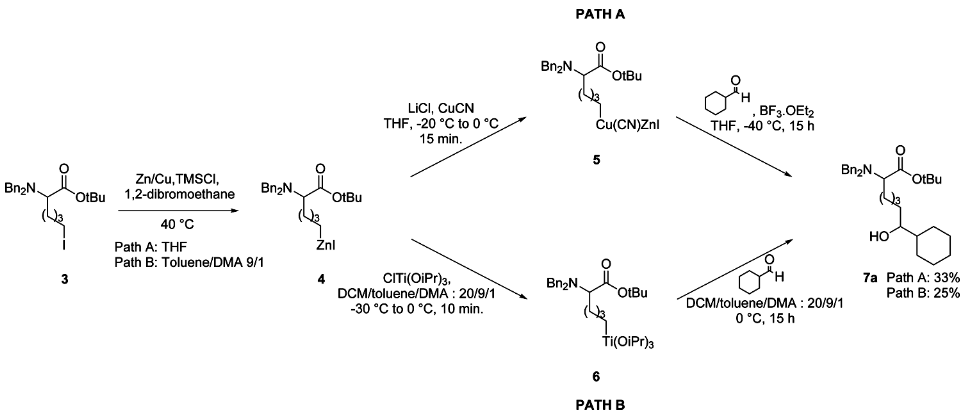
Table 1. Influence of the zinc activation method and the solvent on the formation of organozinc reagent **4**

Activation method of the zinc powder	Solvent	Time (h)	Conversion (%) to **4**
I_2	THF	15	0
1,2-Dibromoethane	THF	15	0
Rieke Zinc	THF	15	44
TMSCl	THF	15	62
Zn/Cu	THF	15	68
Zn/Cu + ultrasound	THF	4	80
1,2-Dibromoethane + TMSCl	THF	2.5	87
Zn/Cu + 1,2-dibromoethane + TMSCl	THF	2	Quant.
Zn/Cu + 1,2-dibromoethane + TMSCl	Et_2O	15	0
Zn/Cu + 1,2-dibromoethane + TMSCl	Toluene/DMA 9/1	2.25	Quant.
Zn/Cu + 1,2-dibromoethane + TMSCl	DMF	1.5	Quant.
Zn/Cu + 1,2-dibromoethane + TMSCl	MeCN	1	Quant.

The use of commercial Rieke zinc[15] as a suspension in THF (Aldrich) was troublesome as it led to large amounts of the corresponding hydroxylated compound, presumably because of the presence of peroxides in the THF suspension. Activation with TMSCl (0.06 equiv.) and Zn/Cu (prepared from zinc powder according to Simmons and Smith[16]) led to **4** in respectively 62% and 68% conversion after 15 h. When Zn/Cu couple was employed under ultrasound activation,[17] the reaction time was dramatically reduced to 4 h with 80% conversion to **4.** Knochel's activation method (TMSCl 0.06 equiv. + 1,2-dibromoethane 0.22 equiv.)[18] on zinc powder led to **4** in 87% conversion in 2.5 h. Combined with the use of Zn/Cu, the reaction time was further decreased to 2 h. Use of diethyl ether with the optimized conditions completely inhibited the reaction. A 9/1 toluene/dimethylacetamide (DMA) mixture[19] gave results similar to THF. Finally, dimethylformamide (DMF) and MeCN led quantitatively to **4** with respective reaction times of 1.5 h and 1 h.

Addition of organozinc **4** on aldehydes was first examined using cyclohexane carbaldehyde as a simplified model for hexose substrates. In early attempts, two transmetalation protocols known to enhance reactivity with aldehydes were investigated. According to Knochel's procedure,[18] **4** was first converted into the zinc/copper reagent **5**, which reacted with cyclohexane carbaldehyde in the presence of $BF_3 \cdot OEt_2$ to give the addition compound **7a** in 33% isolated yield (Scheme 3). In the second protocol, **4** was prepared in a toluene/DMA 9/1 mixture. The organozinc compound was then transferred to a chlorotriisopropoxytitanium(IV) solution in dry dichloromethane (DCM) and cyclohexane carbaldehyde was added to give **7a** after 15 h at 0 °C in 25% yield. In the latter case, the reaction is believed to proceed through the formation of organotitanium **6**.[19a,20]

Direct reaction of **4** on cyclohexane carbaldehyde mediated by $BF_3 \cdot OEt_2$ in MeCN in a one-pot process gave better results (Scheme 4). Preparation of **4** was first performed in MeCN using optimized conditions for zinc activation (Zn/Cu + 1,2-dibromoethane + TMSCl). After cooling the reaction mixture to −40 °C, cyclohexane carbaldehyde (0.5 equiv.) and $BF_3 \cdot OEt_2$ (2 equiv.) were successively added. The reaction mixture was allowed to slowly reach room temperature, and stirring was pursued for 15 h. After aqueous workup and column chromatography isolation, **7a** was obtained in 55% yield.

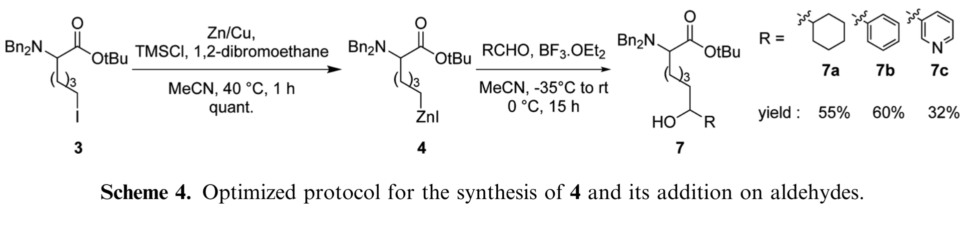


Scheme 3. Synthesis of **7a** from **4** by transmetalation.

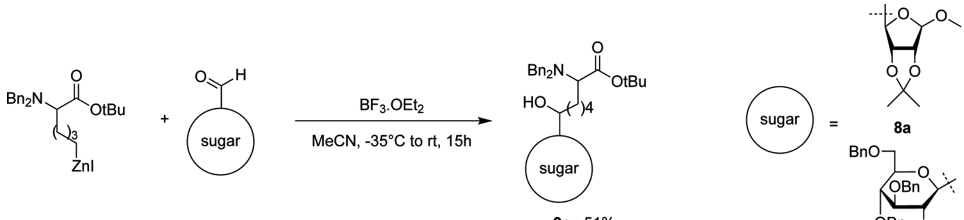
These latter conditions were also tested using aromatic aldehydes (Scheme 4). Benzaldehyde and 2-formylpyridine led to addition products **6b** and **6c** in 60% and 32% yields respectively. 2-Formylthiophene and 2-formylpyrrole failed to give the expected addition compounds but yielded complex reaction mixtures presumably due to the sensitivity of these substrates to $BF_3 \cdot OEt_2$.

We performed the addition of **4** on ribose and glucose derivatives bearing an aldehyde function to prepare unreported *C*-glycosylated α-amino acids (Scheme 5). Riboside **8a** was prepared from D-ribose by simultaneous protection of positions 1, 2, and 3[21] followed by oxidation of the hydroxyl group with iodoxybenzoic acid (IBX).[22] Glucoside **8b** was obtained in four steps from D-glucose according to a procedure published by our group.[23] We used our optimized conditions for preparation of **4** and its direct addition on **8a** and **8b** mediated by $BF_3 \cdot OEt_2$ gave the expected compounds **9a** and **9b** in 51% and 38% yields respectively. Compound **9a** was obtained with 50% diastereoisomeric excess, as determined by ^{1}H NMR analysis at 400 MHz of the crude mixture (signals for protons on position 4′ of the ribose moiety at 4.24 and 4.29 ppm were used for diagnosis). Preparation of compound **9b** did not lead to any diastereoisomeric excess (as determined by ^{1}H NMR analysis at 400 MHz, observing signals for protons on the carbon bearing the hydroxyl function at 3.35 and 3.38 ppm). In both cases, attempts to isolate the different diastereoisomers by column chromatography were unsuccessful, and the compounds were characterized as mixtures of diastereoisomers.

In summary, we developed a simple method for the addition of an amino acid organozinc reagent to various aldehydes. Choosing an organozinc reagent as



Scheme 4. Optimized protocol for the synthesis of **4** and its addition on aldehydes.



4 8

Scheme 5. *C*-glycosylated amino acid synthesis.

nucleophile allowed us to use an amino acid precursor in a commonly protected form, without further modification of its amino and acid functions. This method was used to synthesize three new amino acids and two novel *C*-glycosylated α-amino acid structures in acceptable yields. The four-carbon tether between the carbohydrate to the amino acid moieties could potentially mimic linkage through an asparagine residue. Work is currently carried out to extend this methodology to L-amino acids with different side-chain lengths and amine deactivating protecting groups. These results will be reported in due course.

EXPERIMENTAL

General Methods

Zinc powder was purchased from Strem Chemical Inc. (Newburyport, MA, USA, 99.9% purity, 325 mesh, ref. 93–3060). The Zn/Cu was prepared according to Simmons and Smith and was kept under argon.[16] Anhydrous solvents were obtained by distillation over drying agents: sodium-benzophenone (THF, Et_2O), calcium hydride + phosphorus pentoxide (MeCN), and calcium hydride (DCM). DMA was dried by storing it over 4-Å molecular sieves. ^1H NMR and ^{13}C NMR spectra were recorded on a Brucker AC 400-MHz spectrometer with chemical shift values in ppm relative to residual chloroform (δ_H 7.26 & δ_C 77.2) and C_6D_5H (δ_H 7.16 & δ_C 128.4). Assignments of ^1H NMR and ^{13}C NMR signals were made possible, using correlation spectroscopy (COSY), heteronuclear multiple quantum correlation (HMQC) and distortionless enhancement by polarization transfer (DEPT) experiments. High-resolution mass spectrometry (HRMS) analyses were performed on a FAB (fast atom bombardment) Jeol JSM DX300-SX 102 spectrometer in positive mode (FAB+) using 3-nitrobenzylic alcohol (NBA) or a glycerol/thioglycerol (50/50) mixture (GT) matrices. RP-HPLC-MS analyses were performed on a Waters (Milford, MA, USA) Millenium system coupled with a Platform II Micromass-Waters spectrometer in electrospray positive ionization mode (ES+). A C18 Nucleosil column (Waters, 5 μm, 4.6 mm × 250 mm) was used with mixtures of MeCN and water (+0.1% formic acid) as eluent at a flow of 0.5 mL per min. Melting points were determined on a Gallenkamp digital melting point apparatus and are uncorrected.

tert-Butyl *N,N*-Dibenzylaminoethanoate 2

tert-Butyl bromoacetate (5 mL, 29.87 mmol, 1 equiv.) was added to a dibenzylamine (12.6 mL, 65.72 mmol, 2.2 equiv.) solution in a EtOH/dioxane 5/4 mixture (22 mL). A white precipitate rapidly formed, and the mixture was refluxed for 8 h. After solvent evaporation, the white residue was taken up in diethyl ether, and the white suspension was washed with NaOH 1 N (4×60 mL). The aqueous layers were extracted with diethyl ether (2×100 mL). The combined organic layers were washed with water (50 mL) and brine (50 mL), dried over anhydrous sodium sulfate and concentrated *in vacuo* to yield an oil, which solidified on standing. Compound **2** was purified by successive crystallizations from petroleum ether (5.70 g). Evaporation of the filtrates and purification by column chromatography (petroleum ether/EtOAc 9/1) yielded more **2** (white crystals, total weight: 8.75 g, 94%). Rf (petroleum ether/ DCM 1/1): 0.42; MSHR (FAB+) for $C_{20}H_{26}O_2N$: calculated: 312.1964, found: 312.1977; mp 65–67 °C (lit.[24] 68 °C); ^1H NMR ($CDCl_3$, 400 MHz) δ ppm: 1.45 (s, 9H, tBu), 3.16 (s, 2H, $CH_{2\alpha}$), 3.79 (s, 4H, CH_2-Ph), 7.20–7.41 (m, 10H, $H_{arom.}$); ^{13}C NMR ($CDCl_3$, 100 MHz) δ ppm: 27.20 ($CH_{3\ tBu}$), 53.42 ($CH_{2\alpha}$), 56.59 (CH_2-Ph), 79.72 ($C_{quat\ tBu}$), 125.99, 126.19, 126.98, 127.22, 127.30, 127.66, 127.86 ($C_{arom.}$), 138.05 ($C_{1\ arom.}$), 169.76 (C=O).

n-Butyllithium in Et_2O

A dry, 100-mL, round-bottom flask under argon sweeping was charged with finely cut lithium (1.6 g, 228.57 mmol, 1 equiv.) and anhydrous diethyl ether (33 mL). A solution of *n*-butyl bromide (10.5 mL, 98.29 mmol, 0.4 equiv.) in anhydrous diethyl ether (18 mL) was added dropwise with a dropping funnel. The first 10 drops were added at room temperature, and addition was continued for 6 h at −10 °C. After 15 h reaction at this temperature, the butyllithium solution was titrated with diphenylacetic acid.[25] Concentrations usually ranged between 1.2 M and 1.5 M. The solution could be kept for a few weeks at −20 °C in the tightly sealed round-bottom flask under argon.

tert-Butyl 2-(Dibenzylamino)-6-iodohexanoate 3

Diisopropylamine (2.81 mL, 20.1 mmol, 1.25 equiv.) in dry THF (250 mL) was cooled to −25 °C under argon, and butyllithium (1.2 M in Et_2O, 15.4 mL, 18.5 mmol, 1.15 equiv.) was added. The mixture was left with stirring at −25 °C for 45 min and was cooled to −40 °C. A solution of **2** (5 g, 16.1 mmol, 1 equiv.) in dry THF (5 mL) was added by syringe over 2 min. The reaction mixture took a characteristic deep purple color over a few minutes. After 30 min at −40 °C, the mixture was cooled to −78 °C, and 1,4-diiodobutane (8.47 mL, 6.2 mmol, 4 equiv.) was added. The reaction mixture was allowed to come back to room temperature slowly and was left with stirring for 15 h. The reaction was quenched with saturated aqueous ammonium chloride (100 mL). The two phases were separated, and the aqueous layer was extracted with diethyl ether (2×100 mL). The combined organic layers were washed with 15% aqueous sodium thiosulfate (100 mL) and brine (100 mL), dried over anhydrous sodium sulfate and concentrated in vacuo. Compound **3** was purified by silica-gel column chromatography (petroleum ether/DCM 8/2) and was isolated as a clear yellow oil (7.29 g, 92%). It was stored over copper turnings, under argon, away from light, at −20 °C. Rf (petroleum ether/DCM 85/15): 0.19; MSHR (FAB+) for $C_{24}H_{33}IO_2N$: calculated: 494.1558, found: 494.1556; RMN ^{1}H NMR ($CDCl_3$, 400 MHz) δ ppm: 1.18–1.66 (m, 6H, $H_{\beta+\gamma+\varepsilon}$), 1.49 (s, 9H, tBu), 3.02 (t, 2H, *CH₂*I, $J_{\delta-\varepsilon} = 6.8$ Hz), 3.09 (t, 1H, CH_α, $J_{\alpha-\beta} = 6.0$ Hz), 3.47 (d, 2H, CH_2-Ph, $J = 13.9$ Hz), 4.82 (d, 2H, CH_2-Ph, $J = 13.9$ Hz), 7.11–7.33 (m, 10H, $H_{arom.}$); RMN **^{13}C** ($CDCl_3$, 100 MHz) δ ppm: 7.35 ($CH_{2\ \varepsilon}$), 27,43 ($CH_{2\ \gamma}$), 28.76 ($CH_{2\ \beta}$), 28.92 ($CH_{3\ tBu}$), 33.39 ($CH_{2\ \delta}$), 54.89 (CH_2-Ph), 56.72 (CH_α), 81.46 ($C_{quat.\ tBu}$), 127.41, 128.67, 129.32, 140.18 ($C_{arom.}$), 172.61 (C=O).

Representative Procedure of Optimized Zinc Activation in MeCN, Zincation of 3, and Direct Addition on Aldehydes: *tert*-Butyl 2-(Dibenzylamino)-7-hydroxy-7-cyclohexylheptanoate 7a

Zn/Cu powder (0.239 g, 3.65 mmol, 6 equiv.)[16] was placed under argon sweeping in a dry, 10-mL, round-bottom flask, gently heated under vacuum with a heat gun, and allowed to cool down to room temperature under argon. Dry MeCN (3 mL) was added, followed by 1,2-dibromethane (12 μL, 0.13 mmol, 0.22 equiv.), and the resulting slurry was heated with a heat gun with stirring until reflux and cooled back to room temperature (three times). While still warm, TMSCl (5 μL, 0.04 mmol, 0.06 equiv.) was added. After 15 min, neat **3** (0.3 g, 0.61 mmol, 1 equiv.) was added dropwise by syringe, and the resulting suspension was warmed up to 40 °C. The mixture was stirred at this temperature under an argon atmosphere for 1 h, at which point all of the iodide had been consumed as judged by thin layer chromatography (TLC; petroleum ether/Et_2O 95/5, Rf = 0.5). The mixture was cooled to −40 °C, and cyclohexane carbaldehyde (40 μL, 0.3 mmol, 0.5 equiv.) was added by syringe, followed by $BF_3 \cdot OEt_2$ (154 μL, 1.22 mmol, 2 equiv.). After 15 h stirring, during which time the mixture was allowed to reach room temperature, saturated aqueous NH_4Cl (10 mL) was added and mixture was extracted with EtOAc (3 × 10 mL). The organic phase was washed with brine (10 mL), dried over anhydrous Na_2SO_4, filtered, and concentrated *in vacuo*. The crude product was purified by column chromatography (petroleum ether/Et_2O 8/2) to give **7a** as a colorless oil (0.161 g, 55%). Rf (EP/EtOAc 8/2): 0,41; MSHR (FAB+) for $C_{31}H_{46}NO_3$: calculated: 480.3478, found: 480.3485; ^{1}H NMR ($CDCl_3$, 400 MHz) δ ppm: 1.44 (s, 9H, $CH_{3\ tBu}$), 1.65 + 1.18 (2 m, 19H, $H_{cyclohexyl}$ and $H_{\beta,\gamma,\delta,\varepsilon}$), 3.10 (m, 1H, *CH*OH), 3.46 (d, 2H, CH_2-Ph, $J_{H\text{-}H'} = 13.8$ Hz), 3.87 (d, 2H, CH_2-Ph, $J_{H\text{-}H'} = 13.8$ Hz), 4.16 (d, 1H, H_α, $J_{\alpha-\beta} = 6.3$ Hz), 7.27 (m, 10H, $H_{arom.}$); ^{13}C NMR ($CDCl_3$, 100 MHz) δ ppm: 25.88 ($CH_{2\ 3\ cyclohexyl}$), 26.37 ($CH_{2\ 4\ cyclohexyl}$), 26.48 ($CH_{2\ \gamma}$), 26.95 ($CH_{2\ \delta}$), 28.56 ($CH_{2\ 2\ cyclohexyl}$), 28.89 ($CH_{3\ tBu}$), 29.80 + 29.75 ($CH_{2\ \beta}$), 34.29 + 34.26 ($CH_{2\ \varepsilon}$), 42.66 ($CH_{2\ 1\ cyclohexyl}$), 53.57 (CH_2-Ph), 66.67 (CH_α), 76.69 (CH-OH), 81.37 ($C_{quat.\ tBu}$), 127.28, 128.82, 128.95, 140.36 ($C_{arom.}$), 173.05 (C=O).

tert-Butyl 2-(Dibenzylamino)-7-hydroxy-7-phenylheptanoate 7b

Compound **7b** was prepared from compound **3** on a 4.06-mmol scale according to the representative optimized procedure (*vide supra*). The addition compound was isolated as a colorless oil after silica-gel column chromatography (petroleum ether/Et_2O 8/2, 576 mg, 60%). Rf (petroleum ether/EtOAc 8/2): 0.38; MSHR (FAB+) for $C_{31}H_{46}NO_3$: calculated: 474.3008, found: 474.2997; ^{1}H NMR ($CDCl_3$, 400 MHz) δ ppm: 1.42 (s, 9H, $CH_{3\ tBu}$), 3.48 (d, 2H, CH_2-Ph, $J_{H\text{-}H'} = 13.5$ Hz), 3.79 (d, 2H, CH_2-Ph, $J_{H\text{-}H'} = 13.5$ Hz), 4.22 (s, 0.5H, H_{α},), 4.24 (s, 0.5H, H_{α},), 4.57 (t, 1H, *CH*-OH, $J = 4.2$ Hz), 7.35 (m, 15H, $H_{arom.}$); ^{13}C NMR ($CDCl_3$, 100 MHz) δ ppm: 25.43 (C_{γ}), 26.16 (C_{δ}), 28.74 ($CH_{3\ tBu}$), 29.65 (C_{β}), 35.24 (C_{ε}), 53.42 (CH_2-Ph), 66.45 (C_{α}), 78.02 (CH-OH), 81.48 ($C_{quat.\ tBu}$), 128.26, 128.70, 128.93, 129.28, 129.62, 129.85, 140.34; 144.66 (C_{arom}), 172.75 (C=O).

tert-Butyl 2-(Dibenzylamino)-7-hydroxy-7-(pyridin-2-yl)heptanoate 7c

Compound **7c** was prepared from compound **3** on a 1.01 mmol scale according to the representative optimized procedure (*vide supra*). The addition compound was isolated as a colorless oil after silica gel column chromatography (petroleum ether/Et_2O 8/2, 58 mg, 30%). Rf (EP/EtOAc 8/2): 0,15; MSHR (FAB+) for $C_{30}H_{39}N_2O_3$: calculated: 475.2934, found: 475.2946; RMN ^{1}**H** ($CDCl_3$, 400 MHz) δ ppm: 1.29 (m, 3H, $H_{\delta'+\gamma+\gamma'}$), 1.36 (m, 1H, H_{δ}), 1.45 (s, 9H, $CH_{3\ tBu}$), 1.57 (m, 3H, $H_{\varepsilon'+\beta+\beta'}$), 1.68 (m, 1H, H_{ε}), 3.08 (t, 1H, H_{α}, $J_{\alpha\text{-}\beta} = 7.5$ Hz), 3.46 (d, 2H, $CH_{2\ Benzyl.}$, $J_{H'\text{-}H} = 13.9$ Hz), 3.83 (d, 2H, $CH_{2\ Benzyl.}$, $J_{H\text{-}H'} = 13.9$ Hz), 4.58 (t, 1H, $H_{CH\text{-}OH}$, $J_{H\text{-}\varepsilon} = 4.0$ Hz), 7.12 (m, 2H, $H_{3+5\ Py}$), 7.25 (m, 10H, $H_{arom.\ Benzyl.}$), 7.59 (m, 1H, $H_{4\ Py}$), 8.44 (m, 1H, $H_{6\ Py}$); RMN ^{13}**C** NMR ($CDCl_3$, 100 MHz) δ ppm: 25.10 ($CH_{2\ \gamma}$), 26.11 ($CH_{2\ \delta}$), 28.48 ($CH_{3\ tBu}$), 29.45 ($CH_{2\ \beta}$), 38.50 ($CH_{2\ \varepsilon}$), 54.47 ($CH_{2\ Benzyl.}$), 61.23 (CH_{α}), 72.70 ($CH_{\ C\text{-}OH}$), 80.82 ($C_{quat.\ tBu}$), 120.37 ($C_{3\ Py}$), 122.23 ($C_{5\ Py}$), 126.87 ($C_{arom.\ Bn}$), 128.19 ($C_{arom.\ Bn}$), 128.64 ($C_{arom.\ Bn}$), 136.61 ($C_{4\ Py}$), 139.97 ($C_{1\ arom.\ Bn}$), 148.19 ($C_{6\ Py}$), 162.17 ($C_{2\ Py}$), 172.45 (C=O).

tert-Butyl 2-(Dibenzylamino)-7-(1′-*O*-methyl-2′,3′-*O*-isopropylidene-β-D-ribofuranos-5′-yl)-heptanoate 9a

Compound **9a** was prepared from compound **3** and compound **8a** on a 2.63-mmol scale according to the representative optimized procedure (*vide supra*). The addition compound was isolated as a colorless oil after silica-gel column chromatography (petroleum ether/Et_2O 9/1, 458 mg, 51%, ee 50%). Two diastereoisomers could be distinguished in NMR spectroscopy in C_6D_6 at 400 MHz (^{1}H) and 100 MHz (^{13}C); signals accounting for the minor isomer are underlined. Rf (petroleum ether/EtOAc 8/2): 0.41; MSHR (FAB+) for $C_{33}H_{48}NO_7$: calculated: 570.3431, found: 570.3448; RMN ^{1}**H** (C_6D_6, 400 MHz) δ ppm: 1.07 (s, 3H, $CH_{3\ IP}$), 1.22 (m, 6H, $H_{\gamma+\delta+\varepsilon}$), 1.35 (m, 2H, H_{β}), 1.38 (s, 3H, $CH_{3\ IP}$), 2.84 (s, 3H, $CH_{3\ OMe}$), 3.29 (m,1H, H_{α}), 3.51 (m,1H, $H_{5'}$)), 3.61 (d, 2H, CH_2-Ph., $J = 13.8$ Hz), 3.95 (2 d, 2H, CH_2-Ph, $J = 13.8$ Hz), $\underline{4.24\ (\text{m, 0.33H, H}_{4'})}$, 4.29 (dd, 0.66H, $H_{4'}$, $J = 8.3$ Hz, $J = 2.3$ Hz), $\underline{4.51\ (\text{d, 0.33H, H}_{2'},\ J_{2'\text{-}3'} = 6.0\ \text{Hz})}$, 4.63 (d, 0.66H, $H_{2'}$, $J_{2'\text{-}3'} = 6.0$ Hz), 4.72 (d,0.66H, $H_{3'}$, $J_{2'\text{-}3'} = 6.0$ Hz), $\underline{4.82\ (\text{d, 0.33H, H}_{3'},\ J_{2'\text{-}3'} = 6.0\ \text{Hz})}$, 4.90 (s,1H, $H_{1'}$), 7.32 (m, 10H, $H_{arom.}$); ^{13}C NMR (C_6D_6, 100 MHz) δ ppm: 26.19 ($CH_{2\ \gamma}$), 26.75 ($CH_{2\ \delta}$), 28.58 ($CH_{3\ tBu}$), 30.02 ($CH_{2\ \beta}$), 33.38 ($CH_{2\ \varepsilon}$), 55.25 (CH_2-Ph.), 55.44 ($CH_{3\ OMe}$), 61.75 (CH_{α}), $\underline{72.12\ (\text{CH}_{5'})}$, 72.58 ($CH_{5'}$), 80.39 ($CH_{3'}$), 80.77 ($C_{quat.\ tBu}$), $\underline{83.37\ (\text{CH}_{3'}),\ 86.37\ (\text{CH}_{2'})}$, 86.71 ($CH_{2'}$), $\underline{91.06\ (\text{CH}_{4'})}$, 92.53 ($CH_{4'}$), 110.62

($CH_{1'}$), 112.25 ($C_{quat.\ IP}$), 127.73; 128.05; 129.45; 140.67 ($C_{arom.}$), 172.22 (C=O), 172.28 (C=O).

tert-Butyl 2-(Dibenzylamino)-7-hydroxy-7-(2′,3′,4′,6′-tetra-*O*-benzyl-β-D-glucopyranos-1′-yl)-heptanoate 9b

Compound **9b** was prepared from compound **3** on a 0.709-mmol scale according to the representative optimized procedure (*vide supra*). The addition compound was isolated as a colorless oil after silica gel column chromatography (petroleum ether/Et_2O 9/1, 124 mg, 38%). Rf (petroleum ether/EtOAc 8/2): 0.17; (FAB+) for $C_{59}H_{70}NO_8$: calculated: 920.5101, found: 920.5114; ^1H NMR ($CDCl_3$, 400 MHz) δ ppm: 1.34 (m, 8H, $H_{\beta,\gamma,\delta,\varepsilon}$), 1.38 (s, 9H, $CH_{3\ tBu}$), 3.31 (m, 1H, H_5), 3.35 (m, 0.5H, *CH*-OH), 3,38 (m, 0.5H, *CH*-OH), 3.47 (m, 1H, H_4), 3.51 (m, 1H, H_2), 3.53 (m, 1H, $H_{6'}$), 3.55 (m, 1H, H_6), 3.66 (m, 1H, H_3), 3.92 (m, 12H, $CH_{2\ Benzyl.}$), 4.14 (s, 0.5H, H_{α}), 4.17 (s, 0.5H, H_{α}), 7.21 (m, 30H, $H_{arom.}$); ^{13}C NMR ($CDCl_3$, 100 MHz) δ ppm: 26.39 ($CH_{2\ \varepsilon}$), 26.84 ($CH_{2\ \delta}$), 28.23 ($CH_{3\ tBu}$), 29.61 + 29.72 ($CH_{2\ \beta}$), 34.21 + 34.18 ($CH_{2\ \varepsilon}$), 53.31 (N-CH_2-Ph), 66.47 (CH_{α}), 69.32 ($CH_{2\ 6'}$), 74.02 (O-CH_2-Ph.), 74.38 (O-CH_2-Ph.), 74.51 (O-CH_2-Ph.), 74.68 (O-CH_2-Ph.), 76.32 + 76.56 (*CH*-OH), 76.41 ($CH_{1'}$), 78.84 ($CH_{5'}$), 79.82 ($CH_{2'}$), 81.41 ($C_{quat.\ tBu}$), 82.52 ($CH_{4'}$), 86.87 ($CH_{3'}$), 128.08, 128.14, 128.60, 129.15; 129.49; 129.71; 129.84; 129.96; 130.40; 130.44; 130.57; 130.78; 130.89; 130.95; 130.99; 139.21; 139.61; 139.63; 139.90;140.51 ($C_{arom.}$), 172.84 (C=O).

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